

Worksheet 2: Section 1.2

Linear Equation Applications

Name: _____

Date: _____

Some Basic Things We Should Know

Geometry

- The sum of interior angles in a triangle is 180°

Percentages

- Percentages: $\text{Part} = \text{base} \times \text{rate}$
- Base is always the original value
- Rate (percentage) is in decimal form

Physics / Motion

- $\text{Distance} = \text{Rate} \times \text{Time}$ or $\text{Distance} = \text{Speed} \times \text{Time}$

Problem-Solving Strategy

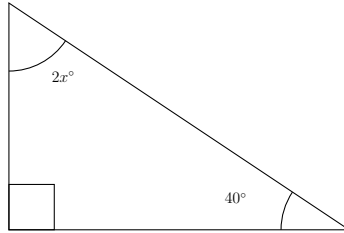
6-Step Problem-Solving Strategy:

1. **Read** – Understand the problem. What are you solving for?
2. **Define** – Assign variables. Figure out how they relate.
3. **Draw** – If possible, draw a picture or graph to visualize.
4. **Equation** – Write an equation using the given information.
5. **Solve** – Solve the equation algebraically.
6. **Check** – Verify your answer works in the original problem.

Practice Problems

Show all work. Box your final answers.

Problem 1. The sum of the measures of the angles in a triangle is 180° . In a right triangle, one angle is 40° , another is $2x^\circ$, and the third angle is a right angle (90°). What is the value of x ?



Solution:

Problem 2. A total of 575 tickets were sold for the school play. They were either adult tickets or student tickets. There were 75 more student tickets sold than adult tickets. How many adult tickets were sold?

Solution:

Problem 3. Farmer John has pigs and chickens. When he looks out into the barnyard he can count 12 animals and together those animals have 30 legs. How many pigs and how many chickens does he have? Solve without an equation.

Solution:

Problem 4. Lisa purchased a prepaid phone card for \$15. Long distance calls cost 16 cents a minute using this card. Lisa used her card only once to make a long distance call. If the remaining credit on her card is \$9.72, how many minutes did her call last?

Solution:

Problem 5. Two passengers leave the airport at Kansas City, Missouri. One flies to Los Angeles, California in 3 hours and the other flies the opposite direction to New York City in 2 hours. With prevailing westerly winds, the speed of the plane to NYC is 60 mph faster than the plane to LA. If the total distance travelled by both planes is 2464 miles, what is the speed of each plane?

Solution:

Problem 6. Tasha considers two sales jobs for two pharmaceutical companies. One pays a base salary of \$25,000 with a 16% commission on sales. The other pays \$30,000 with a 15% commission on sales.

a) Write equations that represent the salaries of the two offers.

b) For how much in sales will the two jobs result in the equal salaries?

c) If Tasha expects to sell \$700,000, which job would be better?

Answer Key

Answer Key

Problem 1: $x = 25$

Problem 2: 250 adult tickets

Problem 3: 3 pigs, 9 chickens

Problem 4: 33 minutes

Problem 5: LA plane: 468.8 mph, NYC plane: 528.8 mph

Problem 6: a) $S_1 = 25000 + 0.16x$, $S_2 = 30000 + 0.15x$

b) \$500,000

c) Job 1 is better (\$137,000 vs \$135,000)